Actuarial Statistics – Confidence Intervals in Health Insurance

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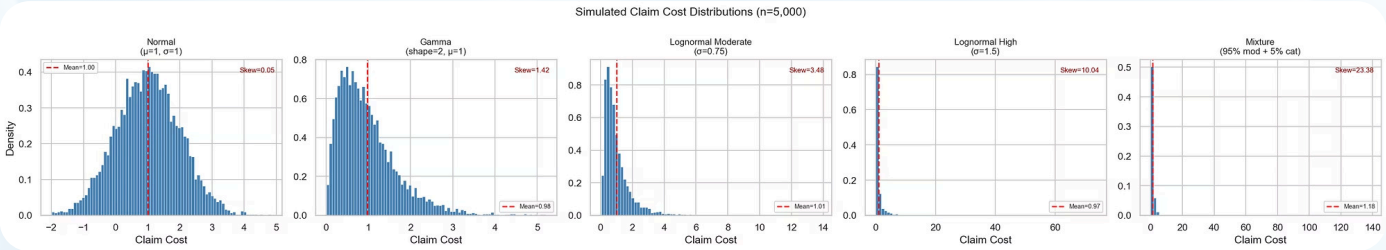
Research Question

Do standard 95% confidence intervals achieve correct coverage for realistic medical claims distributions?

Claim Distribution

Medical claims are right-skewed with rare but extreme "catastrophic" claims. Each distribution may represent a different type of claim groups, from pharmacy claims with gamma to high-risk groups with larger, more often large claim amounts.



- Notice skewness increasing as the chart variability increases. See x-axis values to see how extreme some claims get.

Simulation Setup, Methods & Ideas

Monte Carlo

- 10000 samples per distribution
- Sample sizes from 25 → 500

Bootstrapping

Bootstrapped 499 times per sample – means it resamples the observed data with replacement 499 times to estimate the distribution of the sample mean. Doesn't assume normality

T-interval

Assumes normality, depends on sample size

Z-interval

Critical value always set at 1.96, assumes large enough sample size

Bootstrap

Takes 2.5th and 97.5th percentile values for upper and lower bounds, not symmetric

- Despite these different intervals, the results stay relatively consistent. Between these three, the t-interval will always have the highest coverage, then z-interval, then bootstrap.

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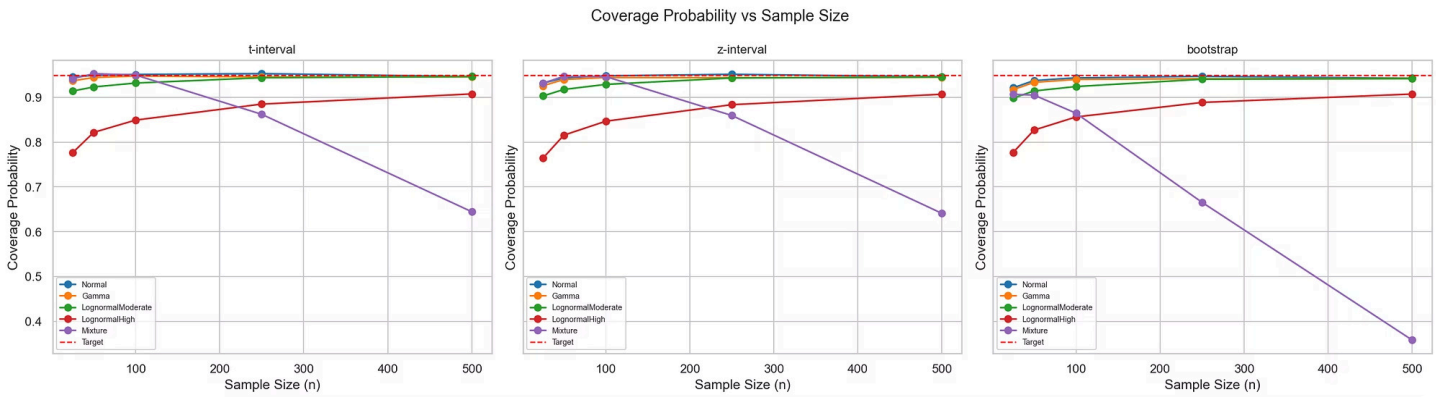
— Snapshot: n=25 vs n=500 (t-interval, 95% CI) —

Distribution	Sample Size	Coverage	Avg Width	Bias	MSE	Type I Error
Gamma	25	0.937	0.5705	9e-05	0.01991	0.063
Gamma	500	0.9456	0.124	-0.00028	0.00101	0.0544
LognormalHigh	25	0.7765	1.5909	-0.00092	0.57254	0.2235
LognormalHigh	500	0.9077	0.4581	0.00082	0.01687	0.0923
LognormalModerate	25	0.9141	0.6678	0.0014	0.02965	0.0859
LognormalModerate	500	0.9466	0.1519	-9e-05	0.00152	0.0534
Mixture	25	0.9416	1.521	0.27117	5.82843	0.0584
Mixture	500	0.6446	0.5876	0.23828	0.159	0.3554
Normal	25	0.9459	0.8167	-0.00276	0.04157	0.0541
Normal	500	0.9466	0.1756	-0.00018	0.00202	0.0534

Only Mixture has significant bias

Increased MSE shows higher variance

Width follows same idea



All distributions except for mixture get closer 95% coverage. This is because for unbiased estimators, CLT says with a large enough sample size, it will follow normal approximation.

Key Metrics

Type 1 Error

Type 1 error = 1 – coverage

Width

Wider intervals increase coverage, but may reduce precision. This is noted for the mixture distribution.

Implications

Undercoverage – actuary will underestimate risk.

Leads to assuming claims cost less, so

- Premiums for members will be underpriced
- Insufficient reserves

Conclusion

> Standard confidence intervals will most likely fail for realistic healthcare claim data, especially under heavy skew and mixture distributions.

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